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Dual Polarized CPW-Fed Multi-Band Antenna for 5G, ISM, and X-Band Applications

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AGENDA

- Abstract
- Introduction
- Design Methodology
- Results
- Conclusion
- References

ABSTRACT

- A coplanar waveguide (CPW)-fed dual-polarized triple-band antenna with an octal-shaped slot and tapering sections in the ground plane is designed for Wi-Fi, Bluetooth, 5G, and X-Band communications, improving impedance matching.

- The antenna achieves circular polarization in the ISM band (2.38-3.56 GHz) with Right-Handed Circular Polarization, while it maintains linear polarization in the 5G band (4.91-5.49 GHz) and X-Band (7.63-8.91 GHz).

- Fractional bandwidths are 46% for the ISM band, 12% for the 5G band, and 16% for the X-Band.

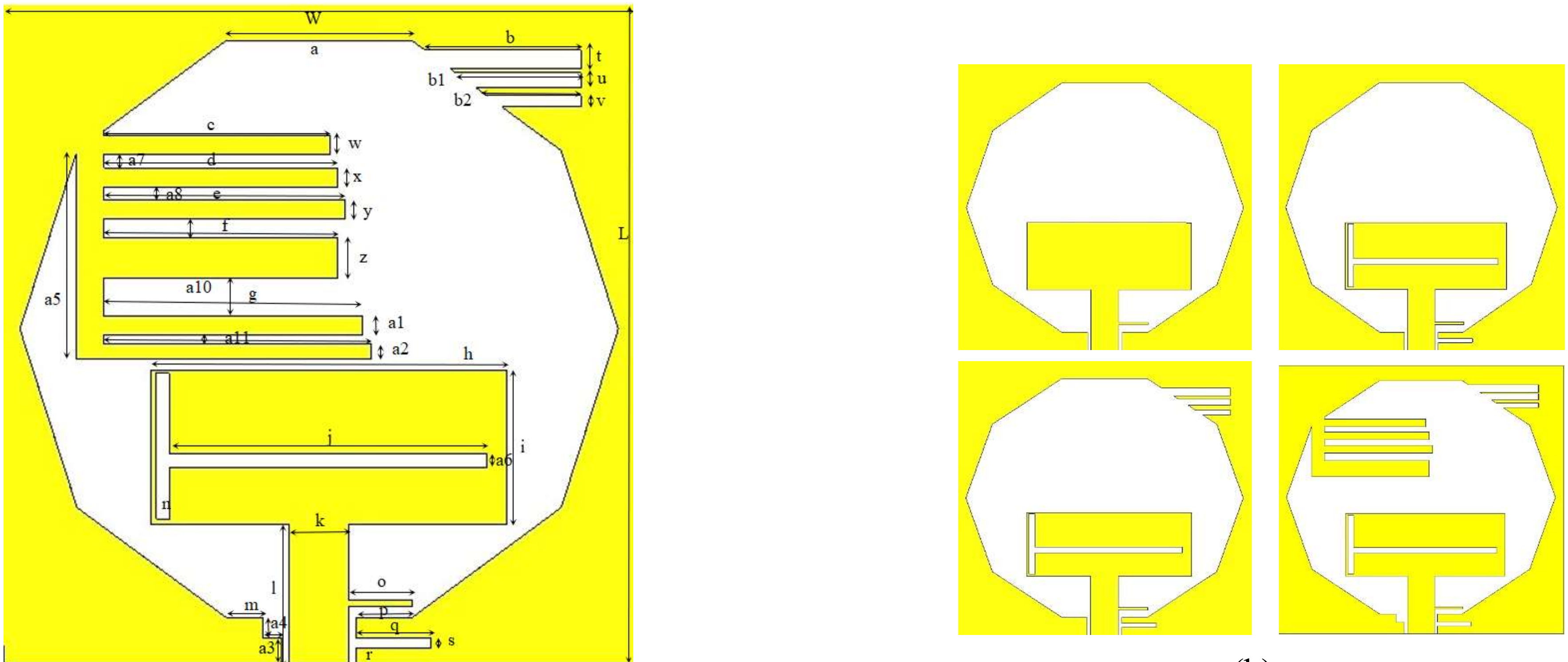
INTRODUCTION

- **Role and Advancements:** Antennas are crucial in wireless communication and radar systems, serving as the interface between electromagnetic waves and electronic devices. Significant advancements have been made in designing compact, wideband, and multiband antennas that maintain performance across various frequencies.
- **Microstrip Antennas:** Microstrip patch antennas are compact, easy to integrate into circuit boards, and offer multiband operation with circular polarization, making them ideal for RF and satellite communications.
- **CPW-Fed Advantages:** CPW-fed antennas use coplanar transmission lines to reduce radiation losses, improve impedance matching, and enhance far-field radiation.
- **Circular polarization:** It is essential for better indoor reception and dynamic device orientations, making these antennas suitable for modern wireless and radar applications.

DESIGN METHODOLOGY

- **Antenna Design and Specifications:** The CPW-fed antenna has dimensions of 34 mm x 35 mm and is built on a Rogers RT-5880 substrate with a thickness of 0.787 mm, a loss tangent of 0.0009, and a relative permittivity of 2.2. It uses a 50 Ω SMA connector for feeding, with the feed line and patch dimensions carefully designed for optimal performance.
- **Benefits of Partial Ground Plane:** The use of a partial ground plane in the design reduces mutual coupling between antenna elements and provides a wider bandwidth compared to a traditional full ground plane.
- **Low Loss Tangent and High Efficiency:** Rogers RT-5880 has a very low loss tangent (0.0009), which minimizes signal loss and enhances the antenna's performance, making it suitable for high-frequency applications.
- **High Dielectric Constant and Compact Design:** With a relative permittivity of 2.2, Rogers RT-5880 supports compact antenna designs while maintaining effective performance characteristics such as gain and bandwidth.

ANTENNA DESIGN



(a)

(b)

Fig1. Proposed Antenna Design (a) and its intermediate design steps (b)

RESULTS

- The simulated plots are obtained as shown in Figure 2. The proposed antenna design operates in three bands which are ISM band (2.38-3.58 GHz), 5G band (4.91-5.51 GHz) and X-Band (7.67-8.93 GHz).

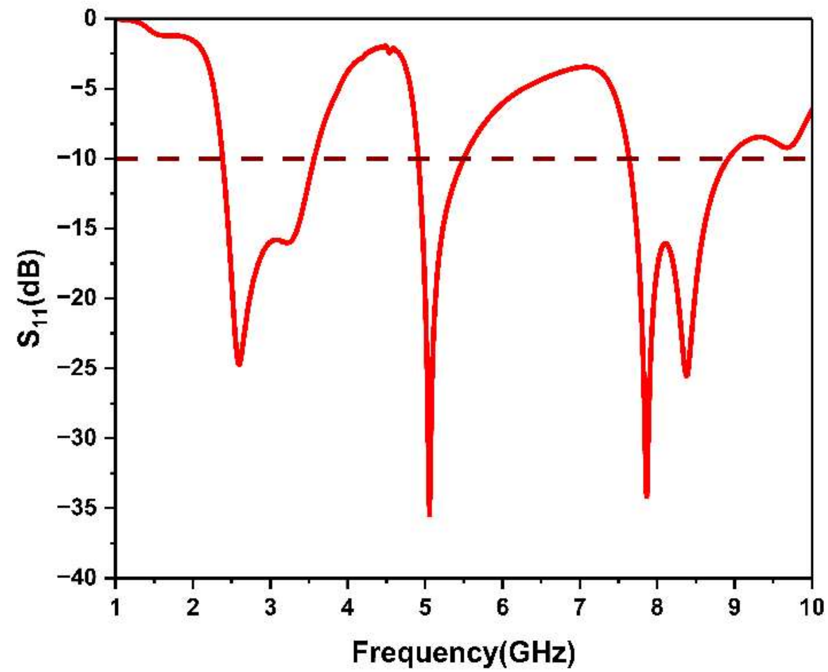


Fig 2. S-Parameter plot

RESULTS

- The Axial ratio of presented antenna can be observed in the s-parameter plot depicted in Fig. 3. It shows the Circular Polarization characteristics in the lower band.
- It can also be observed that the proposed antenna design exhibits a Left-Handed Circular Polarization in the ISM band frequencies while the 5G Band Frequency and the X-Band Frequency only shows Linear Polarization.

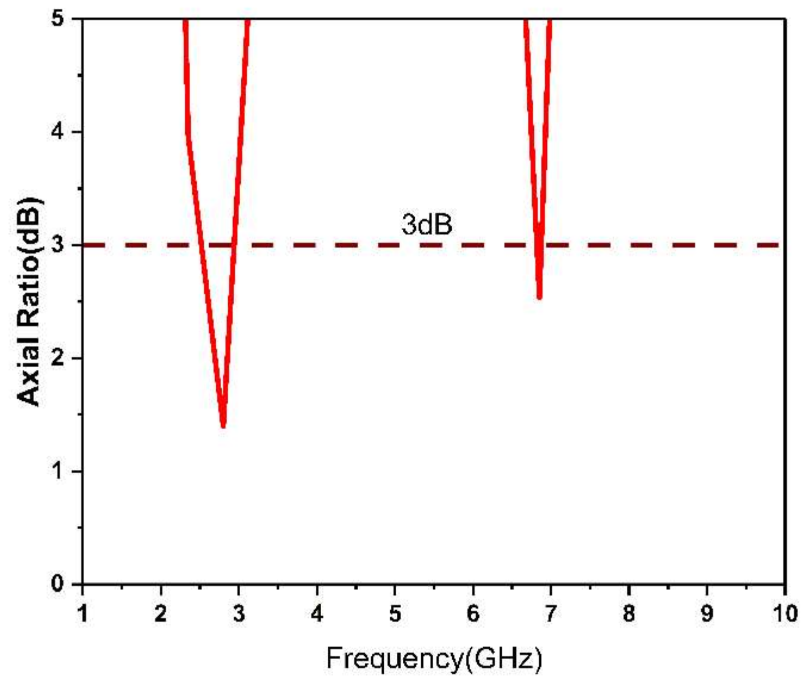


Fig. 3. Axial ratio of the presented antenna

RESULTS

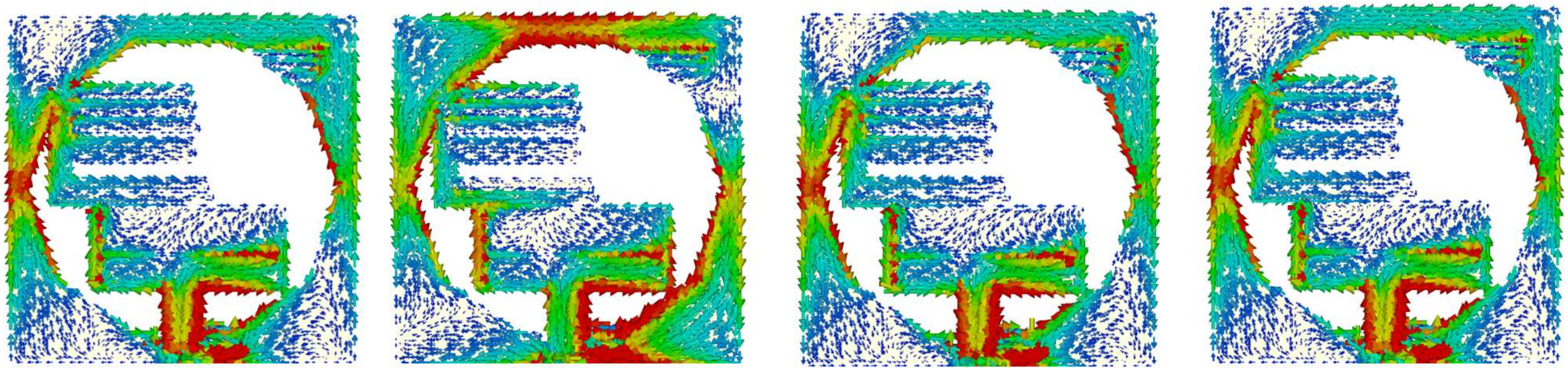


Fig. 4. Surface current distribution of proposed array at 2.4 GHz.
At different phases (a) at 0deg, (b) at 90deg, (c) at 180deg and (d) 360deg.

From the above surface current distributions we can observe that the arrows are rotating counterclockwise. Hence we can conclude that the proposed antenna shows LHCP characteristics at 2.4 GHz.

RESULTS

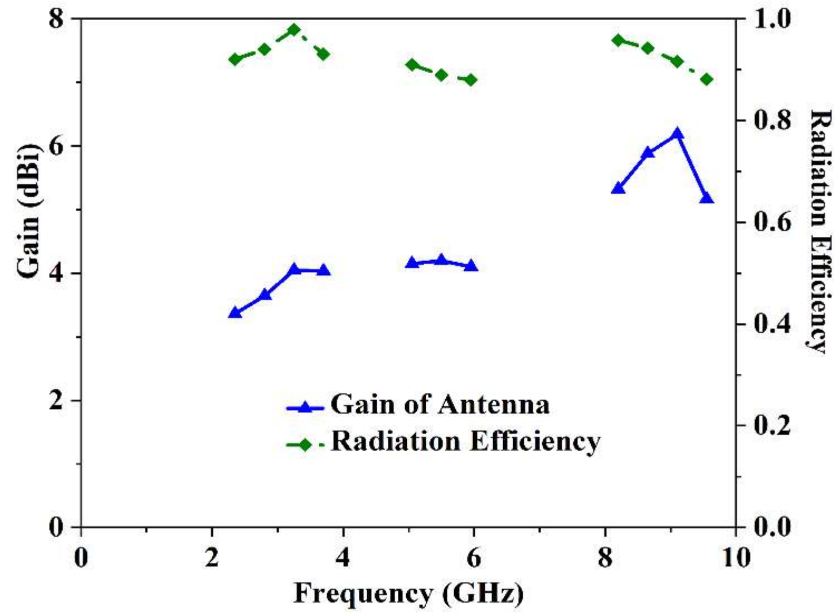


Fig. 5. Realized Gain and effieciency plot of presented antenna.

The gain of the octal shaped CPW-fed antennas array has been sketched in Fig. 5.
From figure, it draws an interferences presented array have gain 3.6 dBi, 4.3 dBi and 6.2 dBi respectively.

CONCLUSION

- **Dual Polarization and Band Coverage:** The antenna is designed to operate over three frequency bands with dual polarization capabilities. It achieves Left-Handed Circular Polarization for the ISM band (covering Wi-Fi and Bluetooth) and Linearly Polarized radiation for the 5G and X-Band communications.
- **Wide Fractional Bandwidth:** The antenna provides significant fractional bandwidths for each frequency range:
    - ISM Band: 46% bandwidth from 2.38 to 3.56 GHz.
    - 5G Band: 12% bandwidth from 4.91 to 5.49 GHz.
    - X-Band: 16% bandwidth from 7.63 to 8.91 GHz.
- **Performance and Gain**: The antenna demonstrates a reasonable realized gain across its return loss bandwidth, maintaining linear polarization for the 5G and X-Band frequencies and circular polarization for the lower ISM band.

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